

Flutter App - Developer Guide

How another engineer can open, edit, run, and build the app.

1. Software Needed

- Flutter SDK
- Android Studio with Android SDK
- VS Code or Android Studio editor
- Git
- Android phone with USB debugging or Android emulator
- Firebase project access

2. App Technology

Item	Details
Language	Dart
Framework	Flutter
Database	Firebase Realtime Database
Main app file	lib/main.dart
Firebase config	lib/firebase_options.dart
Android permissions	android/app/src/main/AndroidManifest.xml

3. Get the Code

Clone the GitHub repository and open the Flutter app folder:

```
git clone https://github.com/YOUR_USERNAME/Smart-Energy-Meter.git
cd Smart-Energy-Meter/Flutter_App
flutter pub get
```

4. Edit the App

- Edit UI, buttons, pages, colors, Firebase paths, and logic in lib/main.dart.
- If new packages are added, edit pubspec.yaml and run flutter pub get.
- Firebase settings are normally in lib/firebase_options.dart.

5. Run on Phone

Connect the Android phone, enable USB debugging, then run:

```
flutter run
```

6. Build APK

Create release APK:

```
flutter clean
flutter pub get
```

```
flutter build apk --release
```

APK output path:

```
build/app/outputs/flutter-apk/app-release.apk
```

7. Build App Bundle for Play Store

```
flutter clean
flutter pub get
flutter build appbundle --release
```

AAB output path:

```
build/app/outputs/bundle/release/app-release.aab
```

8. Important Files

File	Purpose
lib/main.dart	Main Flutter app UI and logic
lib/firebase_options.dart	Firebase configuration
pubspec.yaml	Package dependencies
android/app/src/main/AndroidManifest.xml	Android permissions and network settings
README.md	Developer instructions

9. Privacy Notes

- Do not publish private API keys, tokens, chat IDs, or private database credentials.
- For public GitHub upload, use placeholder values and explain that developers must add their own Firebase project.